🎓 Digital Egypt Pioneers Initiative

Supply Chain
Analysis & Optimization

Project Documentation & Presentation

Team 2 : Nehal Al Sharkawy • Salma Ashraf • Mohammed Ali • Marwa Ali • Hadeer Al Saqa • Hasnaa Amer

Supervised by: Eng. Maged Magdy

4 Weeks	3 Tabs	5 KPIs
Sprint Duration	Dashboard Views	Tracked Metrics

Project Overview

What we built — and why it matters

❗ Problem Statement

▸ Visibility Gaps

No centralized view of supplier performance or logistics costs

▸ Scattered Data

Raw CSV data with no actionable insights or trend analysis

▸ No ROI Clarity

Impossible to identify which routes & suppliers yield the best return

▸ Manual Process

Finance and ops teams relied on manual Excel calculations

✅ Our Solution

▸ Interactive Power BI Dashboard

3-tab navigation: Overview → Supply Chain → Quality & Production

▸ DAX-powered KPIs

Revenue, Profit, Defect Rate, Lead Time — all auto-calculated

▸ Supplier Risk Analysis

Defect rates + Lead times per supplier, cross-referenced

▸ Logistics Optimization

Route & carrier comparison by cost and efficiency metrics

Meet The Team

Digital Egypt Pioneers Initiative • Team 2 • Supervised by Eng. Maged Magdy

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Nehal Al Sharkawy

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Salma Ashraf

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Mohammed Ali

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Hadeer Al Saqa

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Marwa Ali

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Hasnaa Amer

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Eng. Maged Magdy

Project Supervisor & Technical Advisor

Provided technical guidance, reviewed DAX logic, and assessed final dashboard quality.

Project Timeline

4-Week Agile Sprint • From raw data to live dashboard

1

Week 1

Data Preparation

- Data cleaning & normalization
- Handling missing values
- CSV schema design

2

Week 2

Exploratory Analysis

- Sales trend identification
- Revenue driver analysis
- Initial KPI definition

3

Week 3

Deep Dive

- Supplier performance audit
- Logistics bottleneck analysis
- DAX measure development

4

Week 4

Visualization

- Dashboard deployment
- UX testing & iteration
- Final documentation

🎯 Final Deliverables

📊 Interactive Power BI Dashboard

3-tab navigation with cross-filtering slicers

✅ QA Testing Report

Data integrity, DAX accuracy & geographic mapping verified

📄 Full Project Documentation

Requirements, system design, DAX code & test results

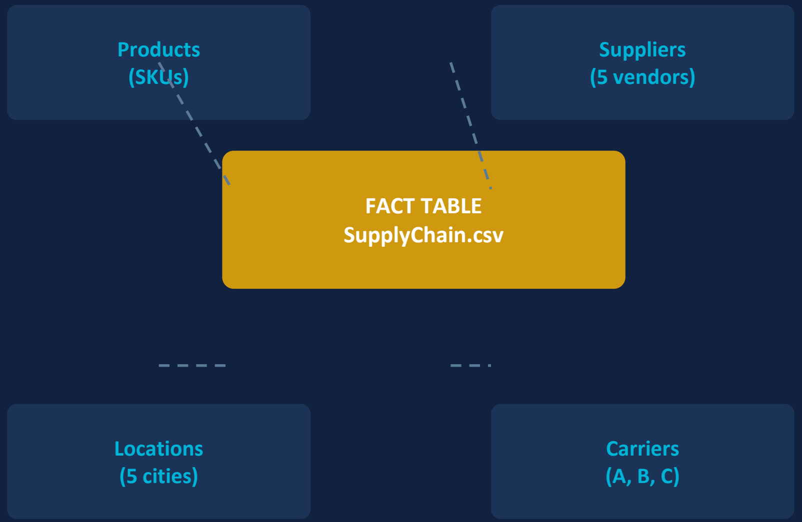
🚀 Strategic Recommendations

+12% profit growth | -5% defect target | +15% shipping speed

System Design & Data Model

Star Schema architecture + DAX logic

⭐ Star Schema Architecture



Primary Key: SKU
Foreign: Product Type → Supplier → Location

⚡ Core DAX Measures

Total Revenue

```
= SUM('SupplyChain'[Revenue_generated])
```

Net Profit

```
= [Total Revenue]
  - (SUM([Mfg_costs]) + SUM([Shipping_costs]))
```

Avg Defect Rate

```
= AVERAGE('SupplyChain'[Defect_rates])
```

Supplier Lead Time

```
= AVERAGE('SupplyChain'[Lead_time])
```

Documentation & QA

How we ensured quality at every stage

📋 Project Planning

KPI targets, 4-week timeline, stakeholder requirements for Ops, Finance & Procurement teams

🏗 System Design

Star schema ERD, UI/UX wireframes, 3-tier dashboard navigation prototype documented

⚙ Implementation

All DAX source code versioned on GitHub with inline comments and measure descriptions

📦 User Manual

Slicer usage guide, cross-filter behavior, drill-through instructions for end users

✅ Test Results — All Passed

✓	**Data Integrity**	Cross-verified with Excel Pivot Tables
✓	**DAX Accuracy**	Manual Net Profit vs. DAX output match
✓	**UX Navigation**	All buttons, page triggers & slicers tested
✓	**Geographic Mapping**	City coordinates mapped to correct locations

🎯 KPI Targets Defined

Financial	Top 20% revenue drivers identified
Logistics	Most cost-effective route identified
Quality	Defect rate maintained below 3%

Outcomes & Recommendations

What the data told us — and what to do next

+12%
Profit Growth Forecasted

–5%
Defect Rate Reduction Target

+15%
Shipping Speed Efficiency

🏭 Prioritize Supplier1 for Skincare

Supplier1 generates $96.7K in Skincare — highest contributor. Lock in longer contracts and increase order share.

⚠️ Review Supplier5 Quality

Highest defect rate at 2.67%. Require corrective action plan and monitor monthly before next procurement cycle.

🚛 Shift Volume to Carrier B

Carrier B is fastest (5.3 days) and cheapest ($5.5). Negotiate preferential rates and increase allocation.

🛤️ Expand Route C Capacity

Route C at $118 is 49% cheaper than Route A. Pilot increased volumes to test capacity limits before scaling.

🚂 Reduce Rail Dependency

Rail = 31% of shipping cost despite not being fastest. Reallocate at least 15% to Road or Sea transport modes.

📦 Address 36% Inspection Fail Rate

Over a third of SKUs fail inspection. Root-cause analysis by supplier is critical before Q3 procurement.

🔗 GitHub: github.com/nehalshark/DEPI-Team2-Supply-Chain-DataAnalysis

SUPPLY CHAIN ANALYTICS

Data Cleaning

Data Cleaning

Step-by-step guide to cleaning supply chain data with practical Python examples

01

Load Data

Load Data

02

Inspect Data

Inspect

03

Clean Data

Clean

04

Validate & Save

Validate & Save

01 & 02 | Load & Inspect Data

1 **Import Libraries**

We import pandas for data processing and os for handling file paths

2 **Set Path & Read File**

We verify the file exists before reading to avoid errors, then load it with pd.read_csv()

3 **First Look at the Data**

df.head() shows the first 5 rows — df.shape tells us the number of rows and columns

4 **Data Types**

df.dtypes reveals if date columns are read as text (object) and need conversion

data_cleaning.py

```python
import pandas as pd
import os

# Set file path
file_path = 'supply_chain_data.csv'

if os.path.exists(file_path):
    df = pd.read_csv(file_path)

    # Inspect the data
    print(df.head())
    print(df.shape)
    print(df.dtypes)

else:
    print('File not found!')
```

03 | Cleaning: Missing Values & Duplicates

Missing Values

df.isnull().sum() → reveals empty values in each column
fillna() → fills gaps with a specified value (zero, mean...)
dropna() → removes rows that contain empty values

Duplicate Rows

df.duplicated().sum() → count of duplicate rows
drop_duplicates() → automatically removes duplicate rows
inplace=True → modifies the DataFrame directly in place

data_cleaning.py — Step 03: Clean

```
# ── Show missing values ──────────────────────────────────
print(df.isnull().sum())

# ── Handle missing numeric values ──────────────────────
num_cols = ['Revenue generated', 'Manufacturing costs', 'Shipping costs']
df[num_cols] = df[num_cols].fillna(df[num_cols].mean())

# ── Handle missing text values ────────────────────────
df['Supplier name'].fillna('Unknown', inplace=True)

# ── Remove duplicate rows ─────────────────────────────────
df.drop_duplicates(inplace=True)
```

04 | Fix Types, Standardize & Validate

1. Fix Data Types

Fix Data Types

Date columns like order_date are read as object → convert with pd.to_datetime() to enable time-series analysis

2. Standardize Text

Standardize Text

str.strip() removes extra whitespace · str.title() standardizes letter casing in columns like Product type

3. Rename Columns

Rename Columns

rename() changes column names to be compatible with Power BI or any visualization tool

data_cleaning.py — Step 04: Fix & Validate

```python
# ── 1. Convert date columns ──────────────────────────────
df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')

# ── 2. Standardize text in product name and supplier columns ──────────
for col in ['Product type', 'Supplier name']:
    df[col] = df[col].str.strip().str.title()

# ── 3. Rename column to standardize the name ────────────────────
df.rename(columns={'Defect rates': 'Defect_Rate'}, inplace=True)

# ── 4. Final validation ──────────────────────────────────
print('Remaining nulls:', df.isnull().sum().sum())
df.to_csv('cleaned_supply_chain_data.csv', index=False)
```

SUPPLY CHAIN

Performance Dashboard

💰 How much are we earning?

📍 Where are we earning it?

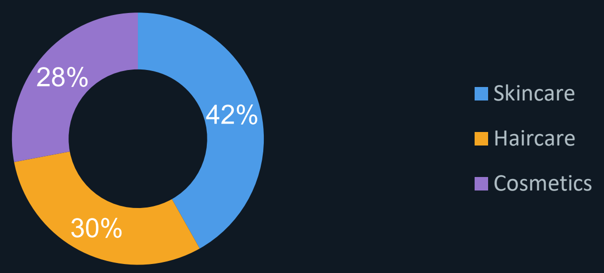
🚀 What products are driving success?

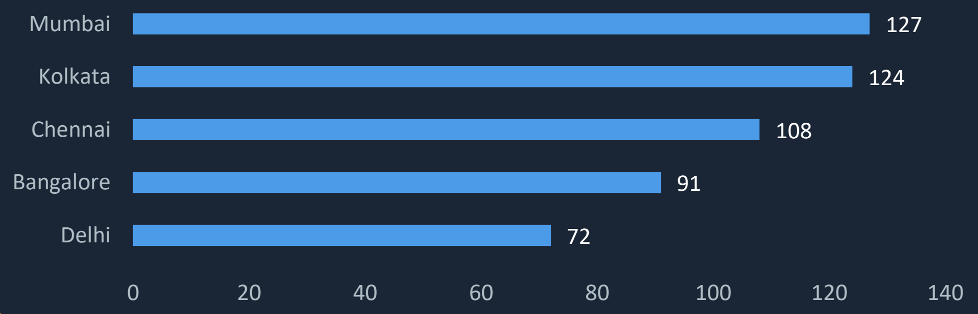
These three questions shaped every design decision in this dashboard.

Financial Overview & Product Mix

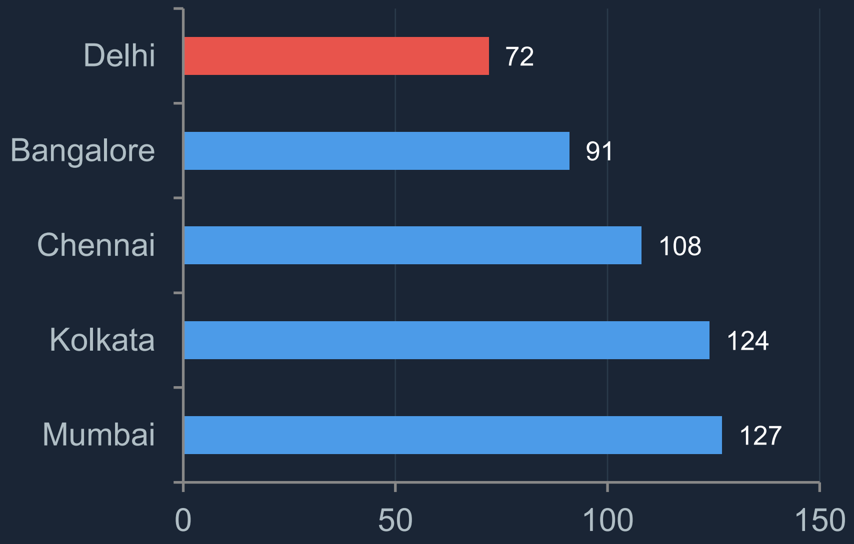
Total Revenue	Actual Profit	Avg Defect Rate	Total Units Sold
$577.6K	$519.4K	2.28%	46K
SUM(Revenue_generated)	SUM(Profit) — Margin: 89.9%	AVERAGE(Defect_rates)	Avg price/unit ≈ $12.5

These KPIs weren't chosen randomly — they cover 3 analytical dimensions simultaneously: **Financial performance, Quality control, and Sales volume.**





Where & What: Regional Profit + Product Efficiency



SKU	Revenue ($)	Turnover
SKU51	9,866	1.54
SKU38	9,862	10.22
SKU31	9,855	28.00
SKU67	9,474	42.75
SKU18	9,364	62.00

Why is Delhi underperforming?

It doesn't mean the market is weak — it means there's a problem worth investigating: higher shipping costs, less efficient suppliers, or stronger local competition. The 3-page dashboard design lets a manager spot the issue in Overview and drill into Supply Chain to investigate.

Multi-Dimensional Analysis

SKU51 has the highest revenue but turnover of only 1.54 — stock is sitting, freezing capital.
SKU31 earns less but turns 28x — fast cash, low risk.
SKU18 turnover of 62 — highest in the portfolio. Best asset for liquidity and cash flow. Understand what drives it and apply it to slower products.

How It Was Built & Key Takeaways

01

DAX Measures

Built all KPI measures in Power BI using SUM, AVERAGE, and calculated columns.

02

Visual Design

Chose each chart type based on the data's nature — not aesthetics.

03

Slicer Integration

Connected all 4 slicers to every visual so any filter affects the full page instantly.

The result: not a dashboard that displays numbers — a dashboard that answers questions.

How much are we earning?

89.9% profit margin

Where are we earning it?

Mumbai & Kolkata lead. Delhi needs attention.

What products drive success?

Skincare dominates. Haircare is the next opportunity.

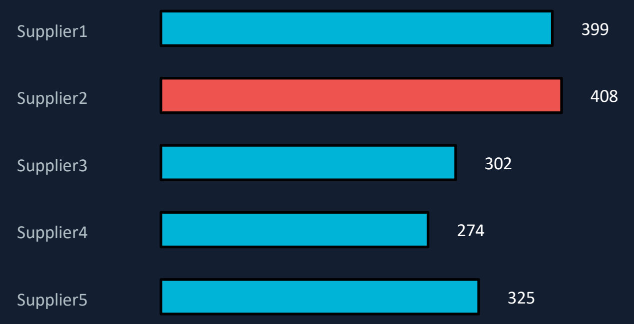
Supply Chain — Dashboard Overview

Supply Chain

Supply Chain Overview Supply Chain Quality & Production

46K	1.01K	5	17
Total Unit Sold	Defects	No. of Suppliers	Avg Lead Time

Lead Time by Supplier



Defect Rates % | Highest: Supplier5 (2.67%) Lowest: Supplier1 (1.80%)

Total Unit Sold — 46K

Represents the total volume of products shipped in the period.
Use this as the denominator when calculating per-unit cost or defect rate.

Total Defects — 1.01K

Total number of defective units recorded across all suppliers.
Benchmark against each supplier's defect rate to find the root cause.

No. of Suppliers — 5

Five active suppliers contributing to the supply chain in this period.
Diversity reduces dependency risk — monitor each supplier's KPIs individually.

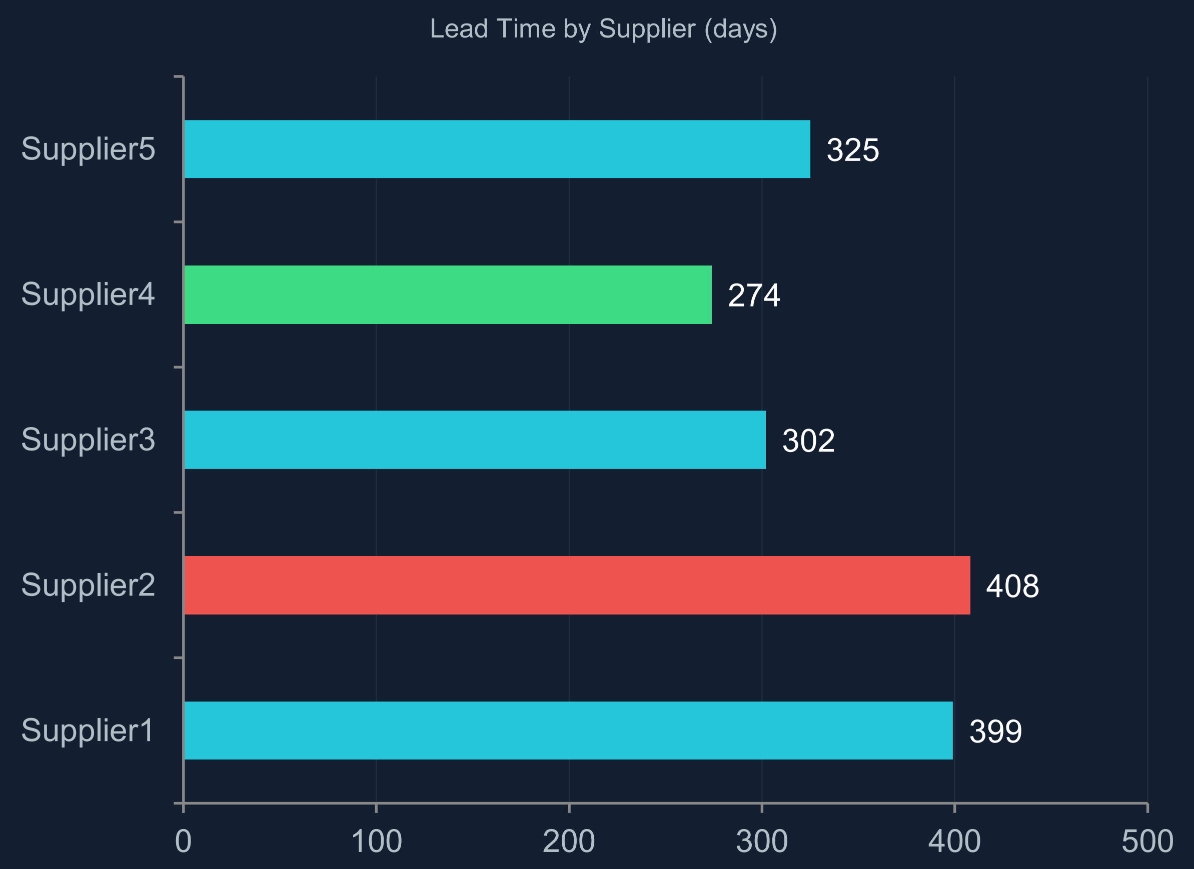
Average Lead Time — 17 days

Average days across all suppliers from order to delivery.
Compare against individual supplier lead times to identify who drags the average up.

🔍 Supplier2 drives the highest lead time (408) — primary optimization target.

Supplier Lead Time Analysis

Supply Chain



What is Lead Time?

Measures how long suppliers take to fulfill and deliver products.
Lower lead times directly improve customer satisfaction and reduce inventory holding costs.
Use this chart to rank suppliers by delivery speed.

Supplier2 — Highest at 408 days

Highest lead time across all suppliers — a primary optimization target.
Investigate whether contract terms, geography, or process inefficiency is the cause.

Supplier4 — Lowest at 274 days

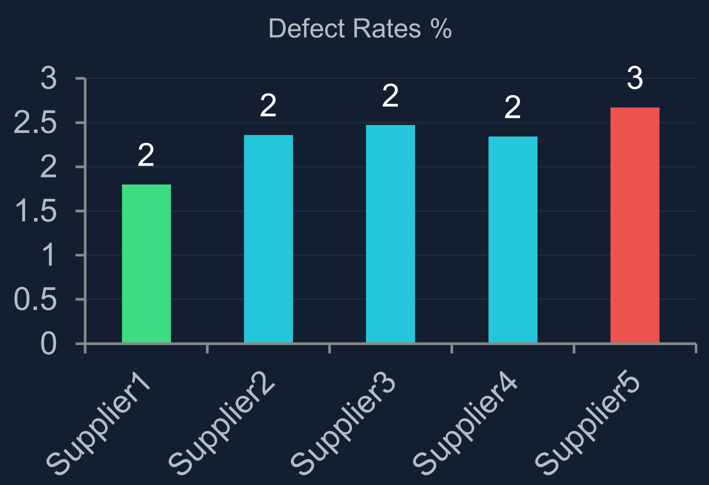
Fastest supplier in the portfolio — benchmark its practices.
Analyze whether Supplier4's process can be replicated by slower suppliers.

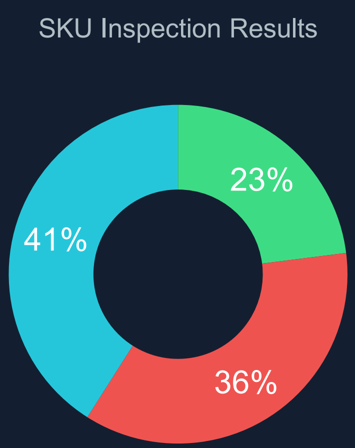
✅ Supplier4 fastest — replicate its practices

⚠️ Supplier2 slowest — review contract immediately

Quality Control and Inspection Results

Supply Chain





Supplier5

Highest Defect 2.67%

Supplier1

Lowest Defect 1.80%

23%

Pass Rate ⚠ Low

Overall defect rate < 3% across all suppliers — quality levels are relatively controlled.

Defect Rates % — Bar Chart

Shows each supplier's defect rate as a percentage of total units.
Supplier5 leads with 2.67% — highest risk to product quality.
Supplier1 is the best performer at 1.80% — use as the quality benchmark.

SKU Inspection Results — Donut Chart

Donut chosen to show part-to-whole split of inspection outcomes.
Pass: 23% | Fail: 36% | Pending: 41%.
The combined Fail + Pending rate of 77% signals a systemic quality control issue.

Key Observation

Pass rate is far below acceptable thresholds.
High pending rate may indicate inspection backlog — prioritize resolution.

⚠ Fail + Pending = 77% — immediate quality review needed

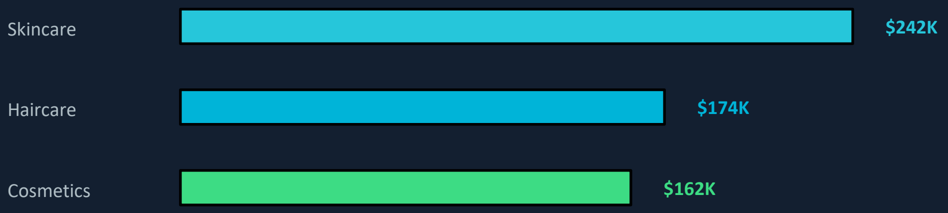
✅ Supplier1 best quality at 1.80% defect rate

Top Supplier Performance Analysis

Supply Chain

Supplier	Cosmetics	Haircare	Skincare	Total
Supplier1	33,228	27,576	**96,728**	157,529
Supplier2	44,576	57,391	23,500	125,457
Supplier3	7,698	28,826	61,472	97,796
Supplier4	25,860	41,918	18,691	86,469
Supplier5	50,160	18,945	41,238	**110,343**
Total	**161,521**	**174,453**	**241,628**	**577,605**

Category Contribution



Suppliers Best Performance Table

Compares each supplier's revenue across Cosmetics, Haircare, Skincare.
Skincare dominates at $241K — highest-value category.
Align sourcing strategy with top-performing product lines.

Supplier5 — Highest Total ($110K)

Top overall contributor despite having the highest defect rate (2.67%).
Prioritize quality improvements for Supplier5 — high volume amplifies defect impact.

Supplier3 — Lowest Total ($97K)

Lowest total contribution across all categories.
Evaluate whether to consolidate its volume with higher-performing suppliers.

Final Recommendations

① Improve quality inspection process — raise pass rate from 23%.
② Reduce Supplier5 defects — high volume amplifies

✅ Supplier5 best revenue performance

⚠️ Supplier3 lowest — review role

Quality & Production — KPI Overview

Quality & Production

46K

Total Unit Sold

$5.55

Ave. Shipping Cost

6

Ave. Shipping Times

4,922

Total Orders

Total Unit Sold — 46K

Represents the total volume of products shipped in the period.
Use this as the denominator when calculating per-unit cost or defect rate.

Ave. Shipping Cost — $5.55

Average cost incurred per shipment across all routes and carriers.
Benchmark this against Route and Carrier breakdowns to spot inefficiencies.

Ave. Shipping Time — 6 days

Average days from dispatch to delivery across all transport modes.
Compare against Carrier Performance chart to identify who drives delays.

Total Orders — 4,922

Total number of orders processed in the analysis window.
Divide by units sold to get average order size — useful for demand planning.

Shipping Cost Analysis

Quality & Production

Donut Chart — Shipping Cost by Transport Mode

Shows proportional cost split: Rail 27.6% | Air 28.2% | Road 28.97% | Sea 15.23%.
Donut chosen because we need to show part-to-whole relationships with the total visible.
Road is the most expensive mode despite not being the fastest — a cost optimization target.

Bar Chart — Shipping Costs by Route

Horizontal bars rank Route A (231) > Route B (205) > Route C (118) by total cost.
Does not show time — must be cross-referenced with Carrier Performance to judge value.
Route C is cheapest: investigate whether it can absorb more volume safely.

⚠ Road = 28.97% of cost — review necessity vs speed benefit

✅ Route C at 118 is 49% cheaper than Route A

Carrier Performance Deep Dive

Quality & Production

Why a Grouped Bar Chart?

We need to compare TWO metrics (Time and Cost) across THREE carriers simultaneously.
Grouped Bar places both bars side by side per carrier — the only chart that shows this clearly.
A line chart or two separate charts would make direct comparison much harder to read.

Carrier Comparison

Carrier A	⏱ 6.14d	$ 5.6	Slowest — review contract
Carrier C	⏱ 6.03d	$ 5.6	Same cost as A, slightly faster
Carrier B	⏱ 5.30d	$ 5.5	Best — fastest & cheapest

✅ Carrier B is the best value — prioritize for high-volume routes

⚠️ Carrier A costs same as C but is 0.11 days slower — no advantage

Logistics Efficiency — Time vs. Cost Analysis

Quality & Production

Why a Scatter Plot?

This is the only chart type that plots TWO continuous variables on X and Y axes simultaneously.
Every dot = one transport mode. Position tells the full story: cheap+fast vs expensive+slow.
No other chart (bar, line, pie) can show this quadrant relationship in a single visual.

How to Read the Quadrants

Bottom-Left
Fast & Cheap
IDEAL — maximize usage

Top-Left
Fast & Expensive
Use only for urgent orders

Bottom-Right
Slow & Cheap
OK for non-urgent bulk shipments

Top-Right
Slow & Expensive
AVOID — worst of both worlds

🔍 Key finding: Sea = bottom-left (best). Rail = top-right (worst). Reduce Rail dependency.

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Supply Chain Analytics
Final Report

Findings • Conclusions • Recommendations

Key Findings

What the data revealed — Part 1 of 2

REVENUE

Skincare Dominates Revenue

Skincare generated ~$241K — 42% of total revenue.
Highest profit margin despite lower unit volume vs. other categories.
Key driver: premium pricing + strong demand in the Indian market.

QUALITY

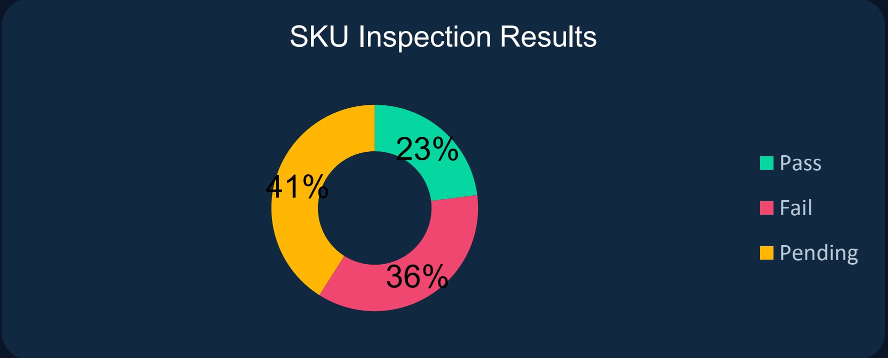
Critical Quality Failure Rate

Only 23% of products PASSED inspection.
36% FAILED outright — 41% still Pending review.
77% of inventory is either defective or unvalidated — a direct threat to brand reputation.

GEOGRAPHY

Mumbai & Kolkata Lead; Delhi Underperforms

Mumbai + Kolkata = ~48% of total revenue combined.
Delhi sells high volume but generates disproportionately low revenue.
Indicates a pricing misalignment or lower-margin product mix in Delhi.



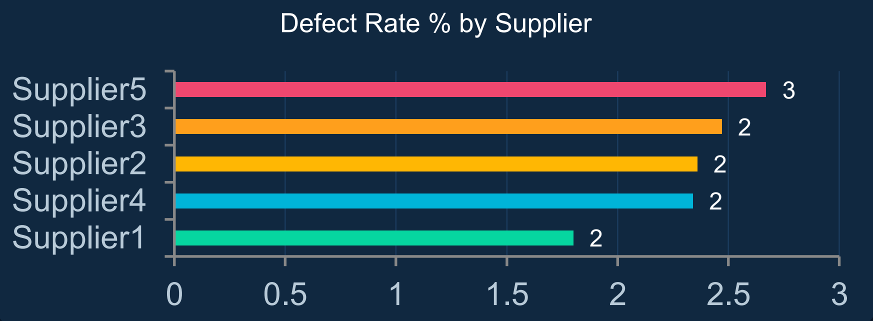
Key Findings

Supplier performance, logistics & predictive analytics — Part 2 of 2

SUPPLIERS

Supplier 5 — Highest Defect Rate

Supplier 5 defect rate: 2.67% — highest across all vendors.
Supplier 1 is the top performer on quality.
Supplier quality is inconsistent — requires continuous KPI monitoring.



LOGISTICS

Transportation Cost vs. Speed Trade-off

Road: fastest avg — 4.7 days. Best for time-sensitive routes.
Air: most expensive — NOT proportionally faster.
Sea: cheapest — significantly slower. Best for bulk non-urgent orders.
Heavy Air usage is cost-inefficient and needs rebalancing.

Prediction

Prediction At-Risk SKUs

We predicted Stock Shortage risk.
Identified 29 SKUs with high probability of running out of stock.
Enables proactive restocking — acting before shortages occur, not after.

Conclusions

Four strategic takeaways from the analysis

1 Financially Strong — but Geographically Concentrated

Profit margin is ~90% — the business is fundamentally healthy.
However, 48% of revenue is tied to just 2 cities (Mumbai + Kolkata).
A demand drop in either city creates significant financial exposure.

2 Quality Control is the #1 Operational Risk

77% of SKUs are either failed or awaiting inspection.
Defective products reaching customers cause returns, loss of trust, and legal risk.
This is a more costly problem than the defects themselves.

3 Inventory Management is Reactive, Not Proactive

Current approach responds to shortages after they occur.
No predictive system was in place before this project.
ML model now provides an early warning system for 29 high-risk SKUs.

4 Data-Driven Decisions Add Measurable Financial Value

Applying analytics + AI can unlock $53K+ in annual savings.
Comes from reduced defect losses, optimized shipping, and better inventory.
This project proves the ROI of investing in a data analytics infrastructure.

Recommendations

Three actionable strategies — with quantified financial impact

R1

Supplier Quality Management

- Conduct a Quality Audit on Supplier 5 immediately
- Reallocate a portion of orders to Supplier 1
- Set monthly KPI reviews on defect rates per vendor
- Define acceptable thresholds — delist non-compliant suppliers

💰 **Estimated Saving: ~$15,000 / year**

R2

Smart Inventory Management

- Define Minimum Stock Levels per SKU
- Integrate ML predictions into weekly restock decisions
- Shift from monthly to weekly inventory reviews
- Prioritize the 29 flagged at-risk SKUs immediately

💰 **Estimated Saving: ~$30,000 / year (avoided lost sales)**

R3

Transport Mode Optimization

- Reduce Air freight share — reallocate to Sea
- Use Sea exclusively for non-urgent bulk shipments
- Audit Route A cost vs. delivery time ROI
- Negotiate better rates with Carrier B (best value carrier)

💰 **Estimated Saving: ~$8,200 / year**

Financial Impact Summary

Current losses vs. projected gains from implementing recommendations

⚠️ Current Annual Losses

▸ Defective Products & Returns	~$18,000
▸ Inefficient Air Freight Overspend	~$12,000
▸ Stockout Lost Sales	~$10,000
▸ Supplier Quality Failures	~$8,000
Total Estimated Annual Loss	**~$48,000**

✅ Projected Annual Savings & Gains

▸ R1 — Supplier Quality Improvement	~$15,000
▸ R2 — Smart Inventory (avoided loss)	~$30,000
▸ R3 — Transport Optimization	~$8,200
▸ Reduced Returns & Brand Protection	~$3,000
Total Projected Annual Return	**~$56,200**

Thank You

Supply Chain Analytics • DEPI Team 2 • 2025

This project demonstrated how Python , Excel and Power BI
can transform raw supply chain data into actionable decisions —
reducing losses, improving quality, and optimizing logistics at scale.

$48K

Current Losses

Identified

$56K

Projected Annual

Savings

29

At-Risk SKUs

Predicted by ML

🔗 github.com/nehalshark/DEPI-Team2-Supply-Chain-DataAnalysis